WebClient Computing

CSS Basics

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Combination of CSS Selectors

- Descendant selector - selectorA selectorB
  - The style applies only to parent-child relation in the document
- Child selector – selectorA > selectorB
  - Applies to only immediate children
- Adjacent sibling – selectorA + selectorB
  - Applies to elements that are only adjacent sibling element
- General sibling – selectorA ~ selectorB
  - Applies to all other sibling elements

Combinator Example

Paragraph 1 in the div.

Paragraph 2 in the div.

Paragraph 3 in the div.

Paragraph 4. Not in a div.

Paragraph 5. Not in a div.

descendant

Paragraph 1 in the div.

Paragraph 2 in the div.

Paragraph 3 in the div.

Paragraph 4. Not in a div.

Paragraph 5. Not in a div.

Adjacent sibling (+)

Paragraph 1 in the div.

Paragraph 2 in the div.

Paragraph 3 in the div.

Paragraph 4. Not in a div.

Paragraph 5. Not in a div.

Child(>)

Paragraph 1 in the div.

Paragraph 2 in the div.

Paragraph 3 in the div.

Paragraph 4. Not in a div.

Paragraph 5. Not in a div.

General sibling (~)

```
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    div p {
      background-color: yellow;
    }
  </style>
</head>
<body>
  <div>
    <p>Paragraph 1 in the div.</p>
    <p>Paragraph 2 in the div.</p>
    <span><p>Paragraph 3 in the div.</p></span>
  </div>
  <p>Paragraph 4. Not in a div.</p>
  <p>Paragraph 5. Not in a div.</p>
</body>
</html>
```

Pseudo Element and Pseudo Class

- Pseudo element - effect to small portion (sub-part) of the element
  - ::first-line
    - Sets an effect to the first line of a block
  - ::first-letter
    - Sets an effect to the first character
  - ::before
    - Sets an effect before the tag happens
- Pseudo class – effect to the entire element
  - :last-child
    - Sets an effects to the last child element

Pseudo Example

THIS IS THE FIRST SENTENSE. THIS IS THE NEXT SENTENCE. THEN A NEXT SENTENCE AND THEN A next sentence

This is a smiley heading

THIS IS THE LAST PARAGRAPH. IT WILL SHOW AN EFFECT OF P:LAST-CHILD WHERE THE EFFECT IS APPLIED ONLY TO THE LAST PARAGRAPH

```
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    p:last-child {
      color: green;
      text-transform: uppercase;
      font-style: italic;
    }
    p::first-letter {
      color: blue;
      font-size: xx-large;
    }
    p::first-line {
      color: red;
      font-variant: small-caps;
    }
    h1::before {
      content: url(smiley.gif);
    }
  </style>
</head>
<body>
  <p>This is the first sentense. This is the next sentence.
  Then a next sentence and then a next sentence</p>
  <h1>This is a smiley heading</h1>
  <p>This is the last paragraph. It will show an effect of
  p:last-child where the effect is applied only to the last paragraph</p>
</body>
</html>
```

Navigation Bar

- Using Lists to organize links

```html
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
</head>
<body>
<ul>
  <li><a href="#home">Home</a></li>
  <li><a href="#news">News</a></li>
  <li><a href="#contact">Contact</a></li>
  <li><a href="#about">About</a></li>
</ul>
</body>
</html>
```

- Home
- News
- Contact
- About

Home

News

Contact

About

```html
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    ul {
      list-style-type: none;
      margin: 0;
      padding: 0;
      width: 200px;
      background-color: #f1f1f1;
    }

    li a {
      display: block;
      color: #000;
      padding: 8px 0 8px 16px;
      text-decoration: none;
    }

    li a:hover {
      background-color: #555;
      color: white;
    }
  </style>
</head>
<body>
<ul>
  <li><a href="#home">Home</a></li>
  <li><a href="#news">News</a></li>
  <li><a href="#contact">Contact</a></li>
  <li><a href="#about">About</a></li>
</ul>
</body>
</html>
```

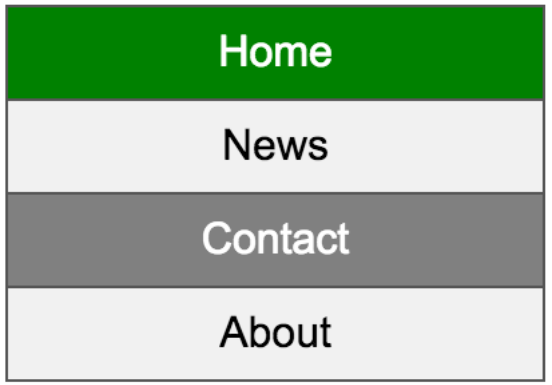
Better Navigation Bar Explanation

- ul list-style-type: none;
  - Do not show bullets
- li a display: block;
  - Originally <a> tag is not a block element but an inline element. Make <a> within <li> as a block element to consume a line
- li a text-decoration: none;
  - By default, a hyper link has an underline and do not use the effect
- li a:hover background-color: #555 and color:white
  - Use a gray color when a mouse is over and change the text color to white
- Margin and Padding in the next slides

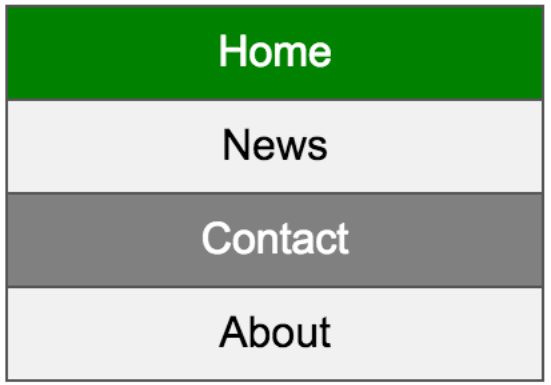
```html
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    ul {
      list-style-type: none;
      margin: 0;
      padding: 0;
      width: 200px;
      background-color: #f1f1f1;
    }

    li a {
      display: block;
      color: #000;
      padding: 8px 0 8px 16px;
      text-decoration: none;
    }

    li a:hover {
      background-color: #555;
      color: white;
    }
  </style>
</head>
<body>
<ul>
  <li><a href="#home">Home</a></li>
  <li><a href="#news">News</a></li>
  <li><a href="#contact">Contact</a></li>
  <li><a href="#about">About</a></li>
</ul>
</body>
</html>
```

Let’s Make This

- Tips
  - There should be outside border
  - Each element in <li> should be aligned center
  - Only the top element (Home) should be in green color
  - The font should be arial with generic type as sans
  - Each element has to have a line underneath (border-bottom)
  - Setting only the last <li> element. Remember pseudo-class of last-child



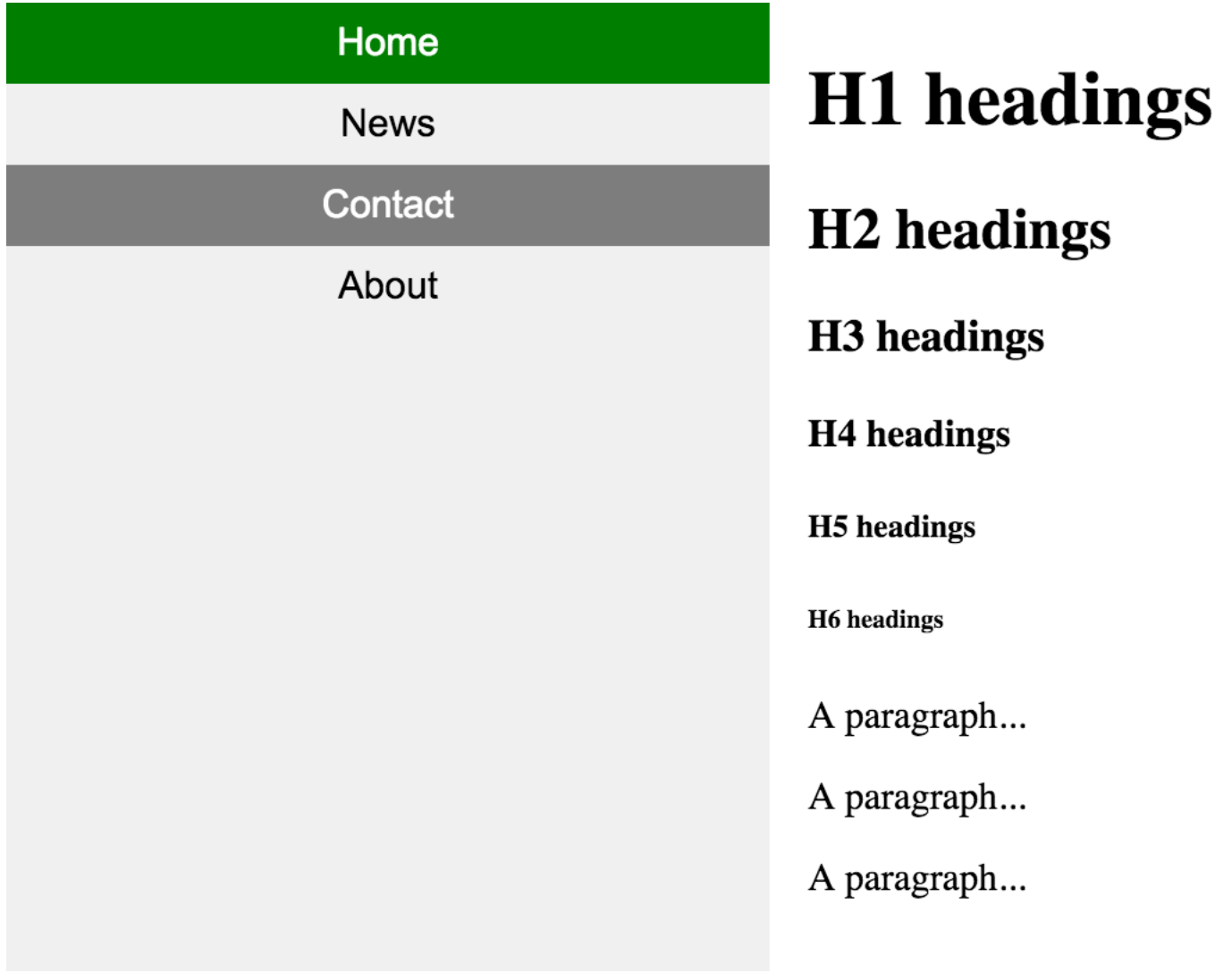
The Outcome




```
ul {
  list-style-type: none;
  margin: 0;
  padding: 0;
  width: 200px;
  background-color: #f1f1f1;
  border: 1px solid #555;
}
li a {
  display: block;
  color: #000;
  padding: 8px 16px;
  text-decoration: none;
}
li a:hover {
  color: white;
  background-color: gray;
}
```

```
li {
  text-align: center;
  border-bottom: 1px solid #555;
  font-family: Arial, sans;
}
li:last-child {
  border-bottom: none;
}
.active {
  background-color: green;
  color: white;
}
```

Fixed Full Height Vertical Navigation Bar



The Code

- margin, padding
  - 1 2 3 4
    - Top
    - Right
    - Bottom
    - Left
  - 1
    - All four dims.
  - 1 2
    - Top and Bottom
    - Left and Right

```
body {
  margin: 0;
}
ul {
  list-style-type: none;
  margin: 0;
  padding: 0;
  width: 25%;
  background-color: #f1f1f1;
  position: fixed;
  height: 100%;
}
li a {
  display: block;
  color: #000;
  padding: 8px 16px;
  text-decoration: none;
}
li a:hover {
  color: white;
  background-color: gray;
}
li {
  text-align: center;
  font-family: Arial, sans;
}
```

```
  .active {
    background-color: green;
    color: white;
  }
  .right-side {
    margin-left: 25%;
    padding: 1px 16px;
    height: 100%;
  }
</style>
</head>
<body>
<ul>
  <li><a class="active" href="#home">Home</a></li>
  <li><a href="#news">News</a></li>
  <li><a href="#contact">Contact</a></li>
  <li><a href="#about">About</a></li>
</ul>
<div class="right-side">
  <h1>H1 headings</h1>
```

How to Make This?

Home

News

Contact

About

H1 headings

H2 headings

H3 headings

H4 headings

H5 headings

H6 headings

A paragraph...

Home

News

Contact

About

The Code

- margin
  - 0
    - All margins 0
  - 0 10
    - Top, bottom 0
    - Left, right 10
  - 0 10 20 30
    - Top 0
    - Right 10
    - Bottom 20
    - Left 30

```css
ul.left {
  list-style-type: none;
  margin: 0;
  padding: 0;
  width: 30%;
  background-color: #f1f1f1;
  position: fixed;
  height: 100%;
}
ul.right {
  list-style-type: none;
  margin: 0 0 0 70%;
  padding: 0;
  width: 30%;
  background-color: #f1f1f1;
  position: fixed;
  height: 100%;
}
```

```html
<ul class="left">
  <li><a class="active" href="#home">Home</a></li>
  <li><a href="#news">News</a></li>
  <li><a href="#contact">Contact</a></li>
  <li><a href="#about">About</a></li>
</ul>
<ul class="right">
  <li><a class="active" href="#home">Home</a></li>
  <li><a href="#news">News</a></li>
  <li><a href="#contact">Contact</a></li>
  <li><a href="#about">About</a></li>
</ul>
<div class="right-side">
  <h1>H1 headings</h1>
  <h2>H2 headings</h2>
```

```css
.right-side {
  margin-left: 30%;
  padding: 1px 16px;
  height: 100%;
}
```

Horizontal Navigation Bar

- By default, <li> tag is a block element and takes the entire width
- Have <li> tag to inline element makes it align in a same line

Home News Contact About

```html
<!DOCTYPE html>
<html>
<head>
<style>
  ul {
    list-style-type: none;
    margin: 0;
    padding: 0;
  }

  li {
    display: inline;
  }
</style>
</head>
<body>

<ul>
  <li><a href="#home">Home</a></li>
  <li><a href="#news">News</a></li>
  <li><a href="#contact">Contact</a></li>
  <li><a href="#about">About</a></li>
</ul>

</body>
</html>
```

Make It Better

Home News Contact About

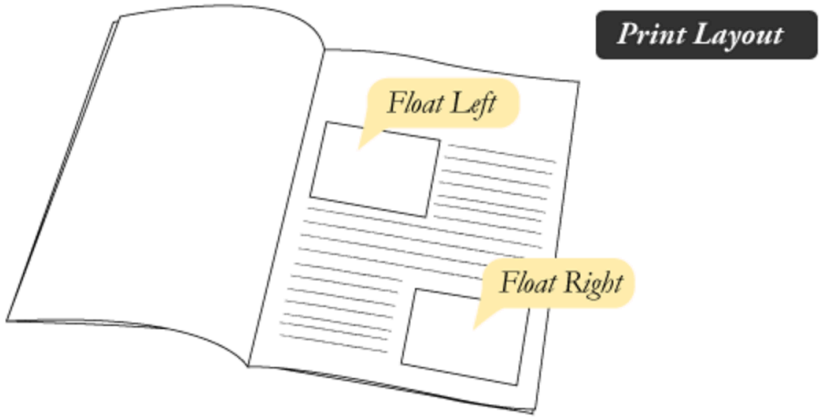
- li - display: inline
  - Have it in horizontal view
- li a - display: inline-block
  - Remember the difference between block and inline elements
  - Have an effect of block in an line
  - Change it to “inline” or ”block”

```html
<!DOCTYPE html>
<html>
<head>
<style>
  ul {
    list-style-type: none;
    margin: 0;
    padding: 0;
    overflow: hidden;
    background-color: #333;
  }
  li {
    display: inline;
    font-family: arial, sans;
  }
  li a {
    display: inline-block;
    color: white;
    text-align: center;
    padding: 14px 16px;
    text-decoration: none;
  }

  li a:hover {
    background-color: #111;
  }
</style>
</head>
<body>
<ul>
  <li><a href="#home">Home</a></li>
  <li><a href="#news">News</a></li>
  <li><a href="#contact">Contact</a></li>
  <li><a href="#about">About</a></li>
</ul>
</body>
</html>
```

Can You Make This?

Home News Contact About

- Tips
  - make a cell goes moves around - float



https://css-tricks.com/all-about-floats/

Locate a Cell Right

```
.right-align {
  float: right;
}
```

```
<ul>
  <li><a href="#home">Home</a></li>
  <li><a href="#news">News</a></li>
  <li><a href="#contact">Contact</a></li>
  <li class="right-align"><a href="#about">About</a></li>
</ul>
```

Home News Contact About

Navigating With the Bar

Home News Contact About

This is H1 heading

This is H2 heading

This is H3 heading

This is H4 heading

This is H5 heading

This is H6 heading

This is H1 heading

This is H1 heading

This is H2 heading

This is H3 heading

This is H4 heading

This is H5 heading

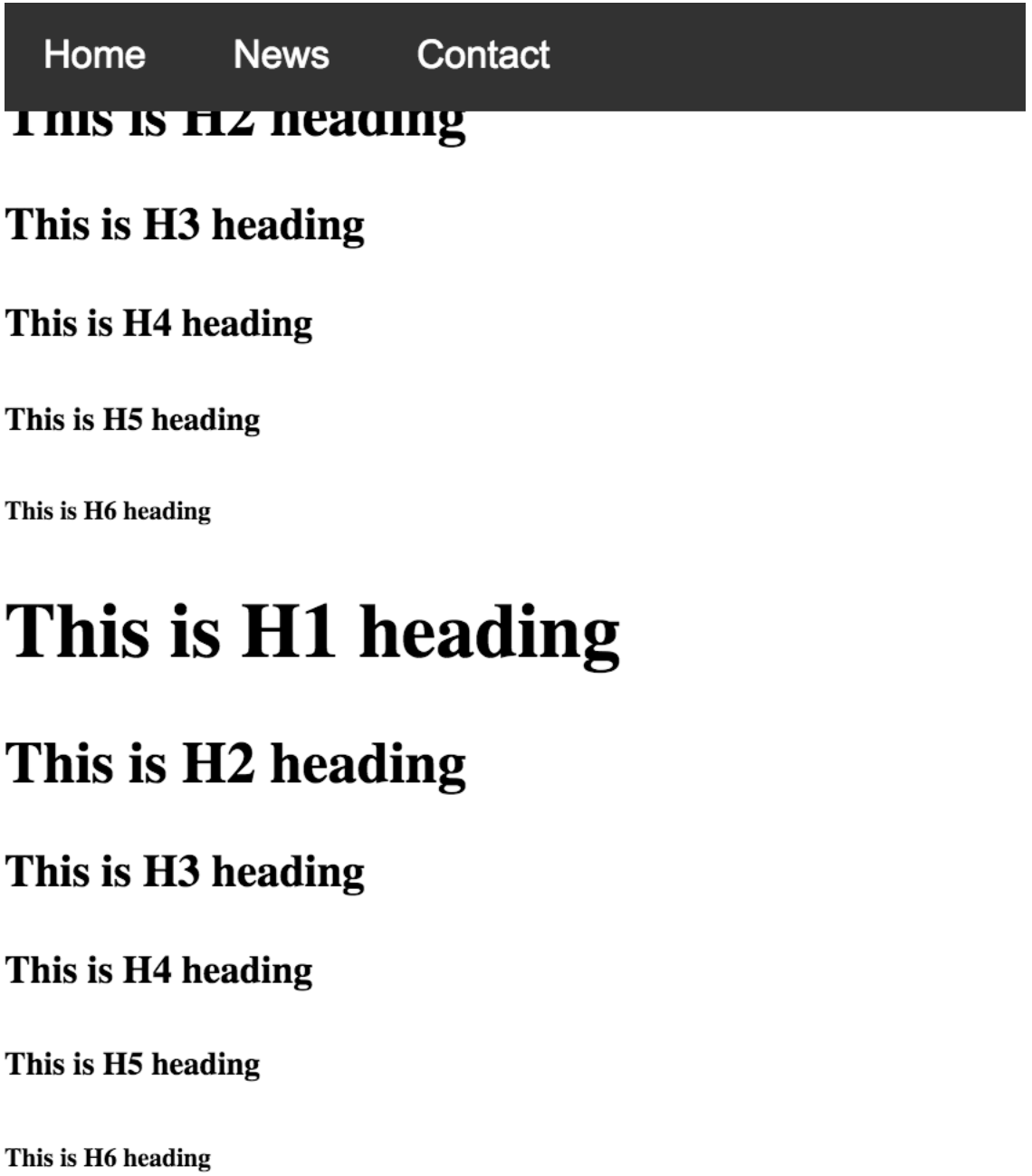
This is H6 heading

This is H1 heading

This is H2 heading

Fixed Horizontal Navigation Bar

```
ul {
  list-style-type: none;
  margin: 0;
  padding: 0;
  background-color: #333;
  position: fixed;
  top: 0;
  width: 100%;
}
```



Reference

- W3 Schools CSS